Evaluating Student Engagement Using Lernavi Data

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I. Introduction

This paper focuses on evaluating student engagement through data from the Lernavi platform. The literature (*J. A. Fredricks, P. Blumenfeld, J. Friedel, A. Paris [1]*) indicates three primary types of engagement: behavioral (*E. L. Deci and R. M. Ryan [2]*), cognitive, and emotional/affective (*K. Salmela-Aro, J. Moeller, B. Schneider, J. Spicer, and J. Lavonen [3]*). Our objective is to identify potential clusters within the available Lernavi data that correspond to these dimensions and then associate these clusters with sutdent's performance.

II. Methodology

A. Engagement Definitions

We first explored student engagement literature in order to have well defined features needed to train our models.

In academic literature [1], engagement is divided into behavioral, emotional, and cognitive dimensions.

Behavioral engagement is defined in several ways :

- Positive conduct, such as following rules and adhering to classroom norms;
- Participation in learning and academic tasks, including behaviors such as persistence, effort or attention;
- Participation in school-related activities.

Cognitive engagement is viewed as a:

- Psychological investment in learning;
- Desire to go beyond requirements;
- Preference for challenge.

Emotional engagement includes students' positive and negative affective reactions in the classroom and toward the school and teacher. It is also conceptualized as identification with the school, including a sense of belonging and valuing success in school-related outcomes.

Table I summarizes the main features extracted from the data and used to cluster types of students according to their engagement style:

TABLE I
Engagement Features Synthesis

Type	Feature & Explanation	Contribution
Behav.	`num_sessions`: Unique sessions / week.	Regularity of student connections on the plaform.
	`event_density`: Total events / week.	Raw activity and overall effort.
	`active_days_per_week`: Active days / week.	Spread of work over time.
	`effort_per_active_day`: Events per active day.	Effort concentration level.
	`usage_regularity`: Inverse of interval std deviation.	Learning routine establishment.
Cognit.	`avg_success_rate`: Avg assessment scores.	Performance and success level.
	`avg_response_latency`: Avg time to answer (sec).	Reflection or perceived difficulty.
	`persistence_score`: Avg attempts before success.	Tenacity and retry willingness.
	`avg_time_to_success`: Time spent on correct answers.	Productive temporal effort.
	`learning_efficiency`: Avg score (0 to 1).	Nuanced view of learning.
	`avg_perseverance_per_topic`: Attempts on failed topics.	Grit on difficult categories.
Emot.	`session_acc / ref`: Count of status.	Attitude towards platform feedback.
	`out_of_school_activity`: Use outside school hours.	Non-imposed engagement.
	`system_adhesion_ratio`: Ratio of accepted sessions.	Trust and error tolerance.
	`intrinsic_motivation_ratio`: Ratio of voluntary acts.	Internal vs External driver.
	`exploratory_curiosity`: View / Click ratio.	Tendency to self-navigate.

B. Clustering models

We used clustering models to explore possible natural links between each type of engagement. We chose to compare two clustering methods: K-Means clustering and spectral clustering. Both methods were found to have an optimal number of 4 clusters. The obtained clusters were then visualized through 2 different dimensionality reduction techniques: PCA and t-SNE. This allowed us to discard spectral clustering for our task, as this method created highly imbalanced clusters, unsuited for any meaningful analysis. However, the visualization of K-Means obtained clusters seem to point in the direction that this method could be suitable to answer our question.

C. Optimal number of clusters

To find the optimal number of clusters, we used different metrics for both clustering methods. For spectral clustering, we

plotted the eigen-values and the silhouette score, while for K-Means clustering, we used the inertia, the silhouette score and the Bayesian Information Criterion (BIC). Combining these metrics led us to conclude that 4 clusters was optimal for both clustering techniques.

III. Results

A. Clustering results

K-means allowed us to find pretty convincing results especially with the PCA visualization technique (Fig 1. It made the results more interpretable compared to t-SNE which might be better suited for larger dimensional objects.

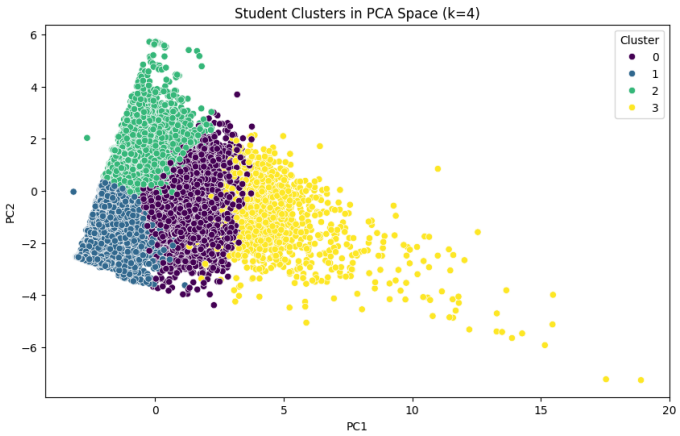

Fig. 1. PCA algorithm on K-means (k=4)

For these clusters we ran mean metric computations for each parameter inside each cluster (Fig 2). This allowed us to judge whether or not the clusters we found were indeed significantly different or, if we had issues with our algorithms or methods. The results seem to be promising as out of the 4 clusters, none seem to be identical to one another and we even have the 3rd cluster really outlying compared to the rest.

Furthermore, we tried to find more intrinsic relationship between the data. Since then we had operated our clustering methods on a merged dataframe containing the metrics used for the 3 types of engagement. We separated the dataframe to have one table per type of engagement and be able to perform more computations; allowing us to find a flow in students behavior and performance. Using a Sankey plot (Fig 3), we were able to showcase how the different students operated between the 3 different categories.

B. Limitations

For the moment, the features concerning cognitive engagement use average over the weeks ; an idea could be to use *Dynamic Time Warping* to be able to work on a weekly success basis in order to figure out differences with the average success ratio.

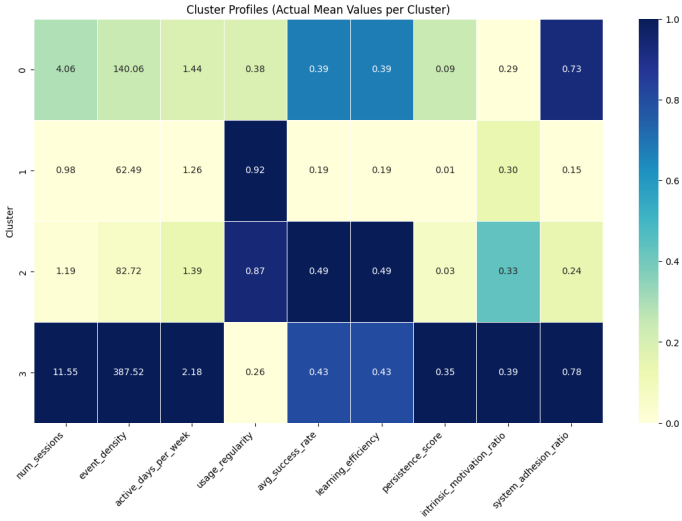

Fig. 2. Cluster profiles

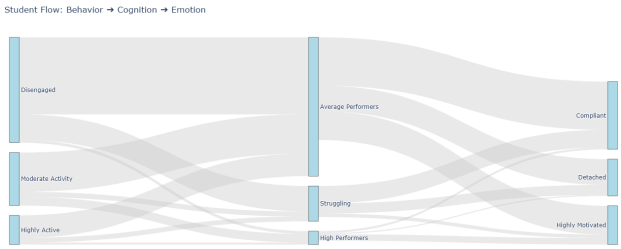

Fig. 3. Flow Chart of the Students

References

[1] J. F. A. P. J. A. Fredricks, P. Blumenfeld, "School engagement," in *Indicators of Positive Development Conference*. Child Trends, mar 2003. [Online]. Available: https://cms.childtrends.org/wp-content/uploads/2013/05/Child_Trends-2003_03_12_PD_PDConfFBFP.pdf

[2] E. L. Deci and R. M. Ryan, "The "what" and "why" of goal pursuits: Human needs and the self-determination of behavior," *Psychological Inquiry*, vol. 11, no. 4, pp. 227–268, 2000. [Online]. Available: https://urls.fr/IQGYaT

[3] K. Salmela-Aro, J. Moeller, B. Schneider, J. Spicer, and J. Lavonen, "Integrating the light and dark sides of student engagement with person-oriented and situation-specific approaches," *Learning and Instruction*, vol. 43, pp. 61–70, 2016. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0959475216300019?via%3Dihub